StatReasoning5

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Activity 13: Statistical reasoning 5: handling complexity using interactions

1. Interactions

Conceptual questions

Q1.1 Describe the interaction in the comic

When the power outage occurs, if you rely on the internet for work, your productivity decreases significantly. If your work does not rely on the internet, post outages, productivity increases slightly to a plateau.

Q1.2 What additional variable would lead to an interaction?

- Bread dough rises because of yeast – the age of the yeast would have an effect on the rate of bread rising. Older yeast would cause the bread to rise more slowly, while fresher yeast would cause the bread to rise more quickly.
- Education leads to higher income – The level of education you receive would lead to differences in income (i.e. PhD would have a higher income than a MS than a BS than a high school diploma…)
- Gasoline makes a car go – The type of car (and therefore the gas mileage) would lead to differences in the amount of gas that would be needed for specific distances.

Penguins

```r
#loading packages
library(tidyverse) # For data wrangling
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(brms) # For stats
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

```r
library(palmerpenguins) # For the data
```

```
Attaching package: 'palmerpenguins'

The following objects are masked from 'package:datasets':

    penguins, penguins_raw
```

```r
library(ggeffects) # for plotting model predictions
```

```r
# Store the data as penguins
penguins <- palmerpenguins::penguins
```

```r
# Look at the column names
penguins %>% colnames()
```

```
[1] "species"           "island"            "bill_length_mm"
[4] "bill_depth_mm"     "flipper_length_mm" "body_mass_g"
[7] "sex"               "year"
```

Q1.3 Filter the dataset to only include Adelie and Chinstrap penguins

```r
penguins.AC <- penguins %>%
  filter(species == "Adelie" | species == "Chinstrap" )
```

Q1.4 Make a plot of flipper length ~ body mass and species

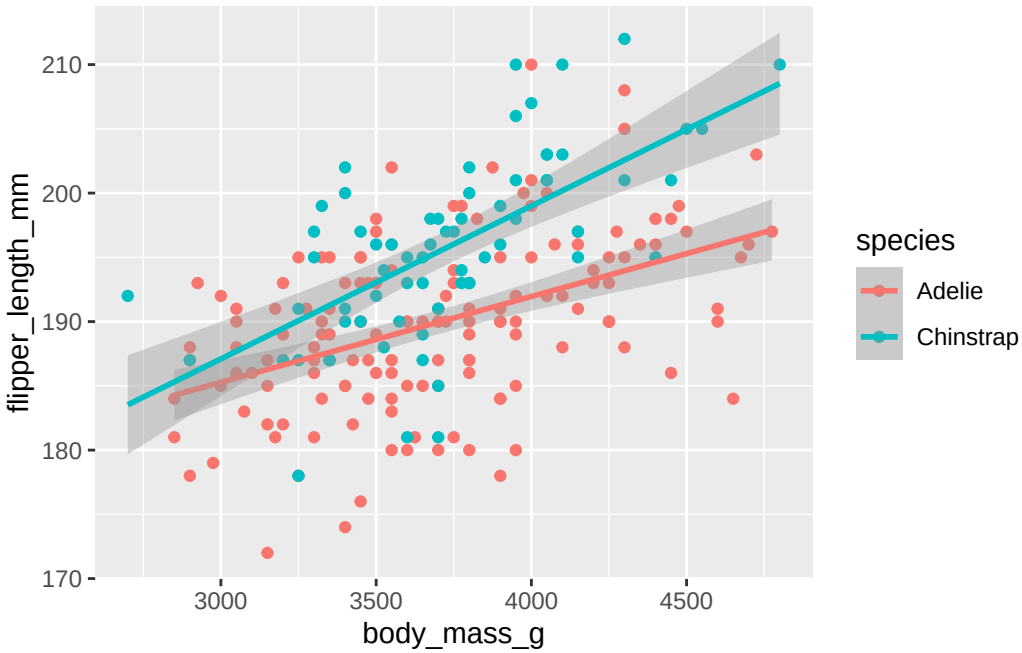
Using ggplot and geom_smooth, make a plot of flipper length on the y axis with body mass on the x, and color the points by species. Put a geom_smooth() line on top of the points with a "linear model" (lm) line.

```r
ggplot(penguins.AC, aes(x= body_mass_g, y = flipper_length_mm, color = species)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

```
Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



Q1.5 Describe the relationships

Do you think that the effect of body mass on flipper length is conditional on species? Why do you think this?

Yes, it seems as if the effect of body mass on flipper length is conditional on species. Although within both species there seems to be a positive effect of body mass on flipper length, Chinstrap have a larger positive effect than Adelie. This makes sense, becuase as individuals with a larger body mass are going to have larger flippers to accommodate their larger size.

Additive model - categorical + continuous

Q1.6 Run and assess an additive model

```
# flipper and body mass model
m.flip.mass.spp.additive <-
  brm(data = penguins.AC, # Give the model the penguin data
      # Choose a gaussian (normal) distribution
```

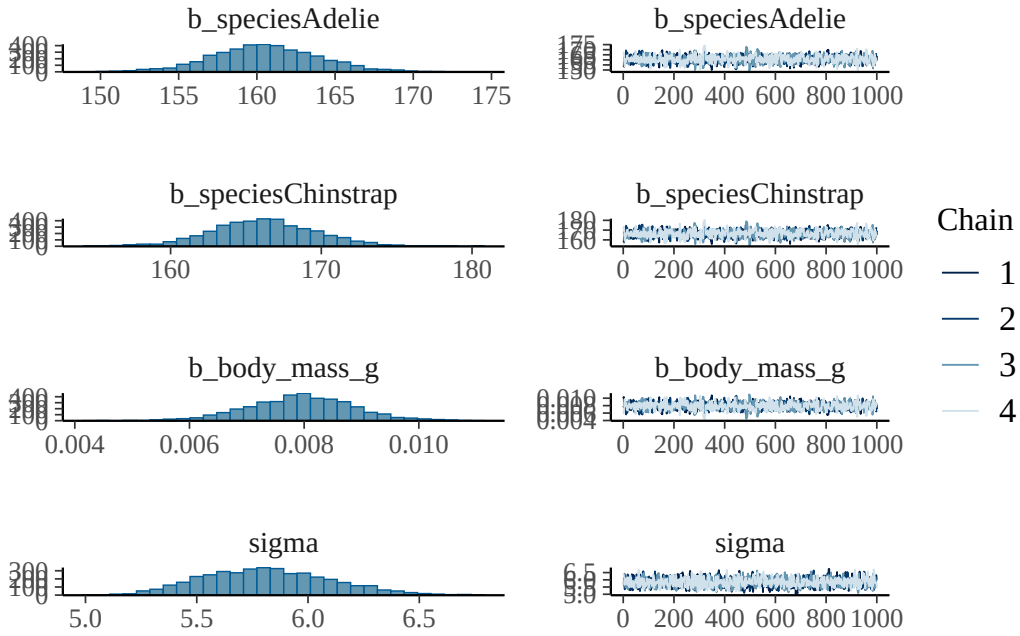
```r
      family = gaussian,
      # Specify the model here.
      flipper_length_mm ~ 0 + species + body_mass_g,
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Setting the "seed" determines which random numbers will get sampled.
      # In this case, it makes the randomness of the Markov chain runs reproducible
      # (so that both of us get the exact same results when running the model)
      seed = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.flip.mass.spp.additive")
```

Summary and plot of model:

```r
summary(m.flip.mass.spp.additive)
```

```
 Family: gaussian
  Links: mu = identity
Formula: flipper_length_mm ~ 0 + species + body_mass_g
   Data: penguins.AC (Number of observations: 219)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
speciesAdelie      160.58      3.44   153.76   167.57 1.01     1164     1320
speciesChinstrap   166.20      3.51   159.22   173.16 1.01     1191     1326
body_mass_g          0.01      0.00     0.01     0.01 1.01     1183     1332

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     5.82      0.29     5.31     6.40 1.00     1538     1633

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

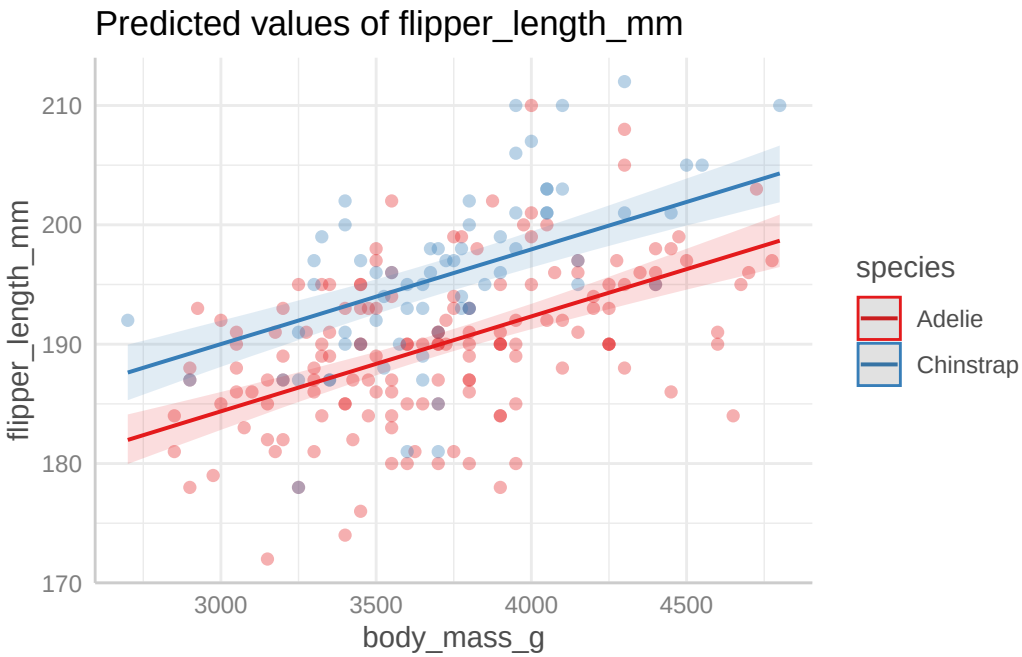
```r
plot(m.flip.mass.spp.additive)
```



Model looks like everything ran well, Rhat is 1, posterior distributions are normal and chains are overlapping. Looks like model ran normally.

```
preds.add <- predict_response(m.flip.mass.spp.additive,
                              terms = c("body_mass_g", "species"))

plot(preds.add, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```



Q1.7 Interpret the additive model

Interpret the additive model output; in a few sentences, answer:

1. Which species has longer flippers, and by how much?

   Chinstrap have longer flippers by 5.62 mm, according to the estimates from the model output.

2. What is the effect of body mass on flipper length (remember units), and is that effect consistent with zero?

   For each gram increase of body mass, flipper length increases by 0.01mm length. The effect is not consistent with zero due to the 95% compatibility intervals being outside of the bounds of zero.

3. Given the additive model, does the effect of body mass on flipper length vary per species?

   It doesn't seem so since the slopes are equivalent between the two species.

```
#To print out more decimal places in the model output
print(m.flip.mass.spp.additive, digits = 4)
```

```
 Family: gaussian
  Links: mu = identity
Formula: flipper_length_mm ~ 0 + species + body_mass_g
   Data: penguins.AC (Number of observations: 219)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                 Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
speciesAdelie    160.5778    3.4416 153.7570 167.5652 1.0073     1164     1320
speciesChinstrap 166.2003    3.5091 159.2169 173.1608 1.0087     1191     1326
body_mass_g        0.0079    0.0009   0.0061   0.0098 1.0081     1183     1332

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sigma   5.8177    0.2857   5.3071   6.4033 1.0028     1538     1633

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Interactive model - categorical + continuous

Now let's try an interactive model! As in the lecture, we are going to write this with "non-linear syntax" within the `bf()` function where our model normally goes.

```r
# flipper length by body mass and species - INTERACTIVE model
m.flip.mass.spp.interactive <-
  brm(data = penguins.AC, # Give the model the penguins data
      # Choose a gaussian (normal) distribution
      family = gaussian,
      # Specify the model here.
      # First, we write the equation with a as an intercept and b as a slope next to body mas
      bf(flipper_length_mm ~ 0 + a + b*body_mass_g,
         # Then, we specify that we want our intercept, a, to vary with species
         a ~ 0 + species,
         # Next, we specify that we want our slope, b, to ALSO vary with species
         # (this is the interaction part!!)
         b ~ 0 + species,
         # Lastly, we tell it that we are writing in a particular notation
```

```r
         nl = TRUE),
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Setting the "seed" determines which random numbers will get sampled.
      # In this case, it makes the randomness of the Markov chain runs reproducible
      # (so that both of us get the exact same results when running the model)
      seed = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.flip.mass.spp.interactive")
```

```r
# Print out the model output with 4 digits to avoid the display rounding to zero
print(m.flip.mass.spp.interactive, digits = 4)
```

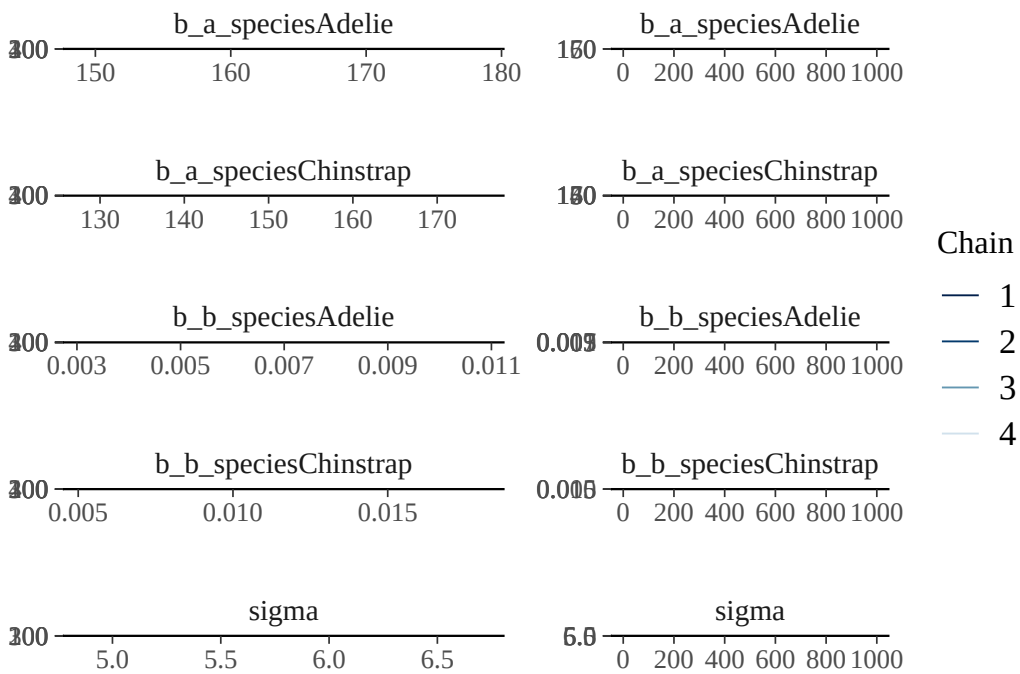
```
 Family: gaussian
  Links: mu = identity
Formula: flipper_length_mm ~ 0 + a + b * body_mass_g
         a ~ 0 + species
         b ~ 0 + species
   Data: penguins.AC (Number of observations: 219)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                   Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS
a_speciesAdelie    165.1809    3.8183 157.5669 172.5782 1.0001     2447
a_speciesChinstrap 151.1712    6.5569 138.2504 164.1750 1.0019     1698
b_speciesAdelie      0.0067    0.0010   0.0047   0.0087 1.0001     2465
b_speciesChinstrap   0.0120    0.0018   0.0085   0.0155 1.0018     1697
                   Tail_ESS
a_speciesAdelie        2131
a_speciesChinstrap     1906
b_speciesAdelie        2124
b_speciesChinstrap     1916

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI   Rhat Bulk_ESS Tail_ESS
sigma   5.7306    0.2723   5.2303   6.3094 1.0016     2133     2075

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

We now have four terms in our model: - an intercept (`a`) for Adelie (165.2524) - an intercept (`a`) for Chinstrap (151.0668) - a slope (`b`) for Adelie (0.0067) - a slope (`b`) for Chinstrap (0.0120)

It looks like both of our slopes are different from zero; the flipper length-body mass relationship for Adelie is 0.0067mm/g (95%CI from 0.0047-0.0086) while for Chinstrap it's nearly doubled at 0.0120mm/g (95%CI from 0.0084-0.0158).

Plot posteriors

```
plot(m.flip.mass.spp.interactive)
```

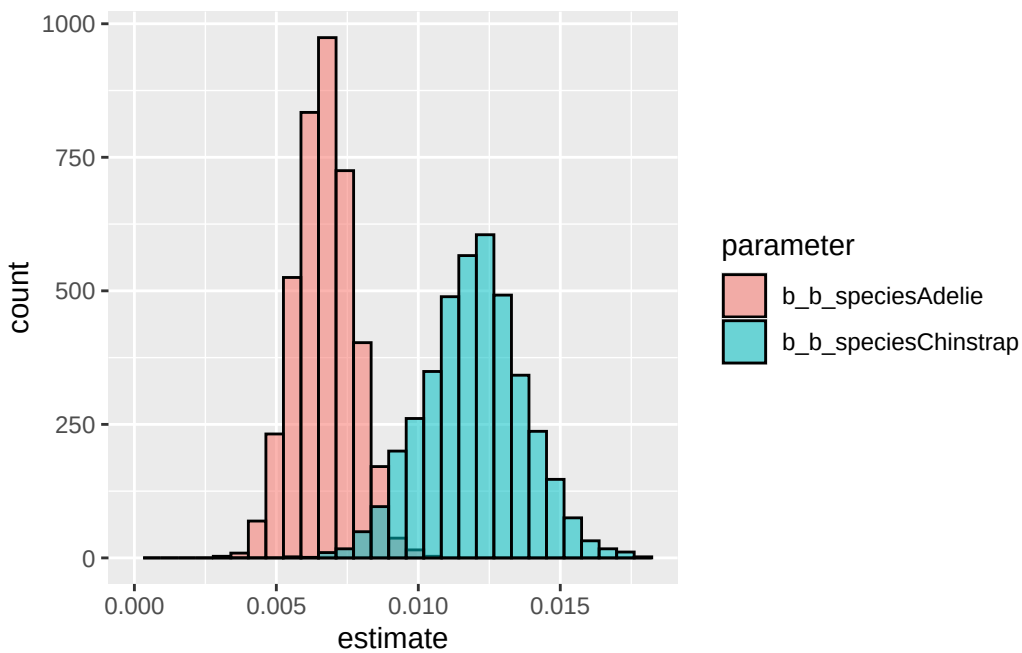


However, it's hard to tell how much the distributions for the two slopes overlap though, because the scales are different. Here's some fun code to extract the raw posterior values that went into this plot and make our own plot with the estimates side-by-side:

```
# Store posterior parameter estimates
model_posterior_samples <- as_draws_df(m.flip.mass.spp.interactive)

# Wrangle and then plot samples for the slopes (b) specifically
model_posterior_samples %>%
  # Select the columns we care about (exclude the posterior estimates for the intercepts and
  dplyr::select(b_b_speciesAdelie, b_b_speciesChinstrap) %>%
```

```r
# Pivot to long format for easy ggplotting
pivot_longer(cols = everything(),
             names_to = "parameter",
             values_to = "estimate") %>%
# Plot as a two overlapping histograms, with the fill colored by parameter
ggplot(aes(x = estimate, fill = parameter)) +
geom_histogram(alpha = 0.55, color = "black",
               # position = "identity" makes the histograms overlay each other instead of s
               position = "identity") +
xlim(0, NA)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
```

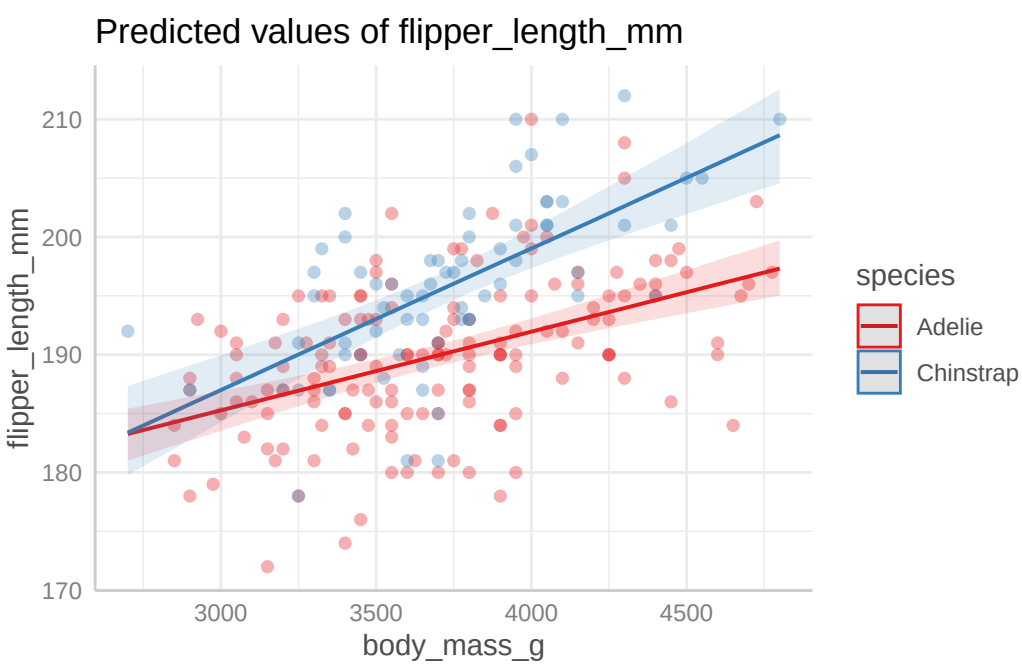


There is only a bit each distribution that overlap, so I feel confident that the estimates are different, and thus we have two different slopes.

Plot model predictions

```
preds.int <- predict_response(m.flip.mass.spp.interactive,
                              terms = c("body_mass_g", "species"))

plot(preds.int, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```



Q1.8 Which model has better predictive power?

Compare the additive vs interactive model using WAIC and PSIS values.

First PSIS:

```
#PSIS additive model
loo(m.flip.mass.spp.additive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

         Estimate   SE
elpd_loo   -697.6 11.1
p_loo         4.0  0.5
looic      1395.3 22.1
------
MCSE of elpd_loo is 0.0.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.3, 1.0]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
#PSIS interactive model
loo(m.flip.mass.spp.interactive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

         Estimate   SE
elpd_loo   -695.3 11.2
p_loo         4.8  0.6
looic      1390.6 22.4
------
MCSE of elpd_loo is 0.0.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.0]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

Then WAIC:

```r
#WAIC addivite model
waic(m.flip.mass.spp.additive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

          Estimate   SE
elpd_waic   -697.6 11.1
```

```
p_waic         4.0  0.5
waic        1395.3 22.1
```

```r
#WAIC Interactive model
waic(m.flip.mass.spp.interactive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

          Estimate   SE
elpd_waic   -695.3 11.2
p_waic         4.8  0.6
waic        1390.5 22.4
```

For both WAIC and PSIS, the interactive model has a lower WAIC and PSIS estimate, confirming that this model has the higher predicitve power.

2. Do it yourself

Now it's your turn to run an *interactive* model. Using the `penguins.AC` filtered dataset, model how `bill_length_mm` varies as a function of `bill_depth_mm` interacting with `species`.

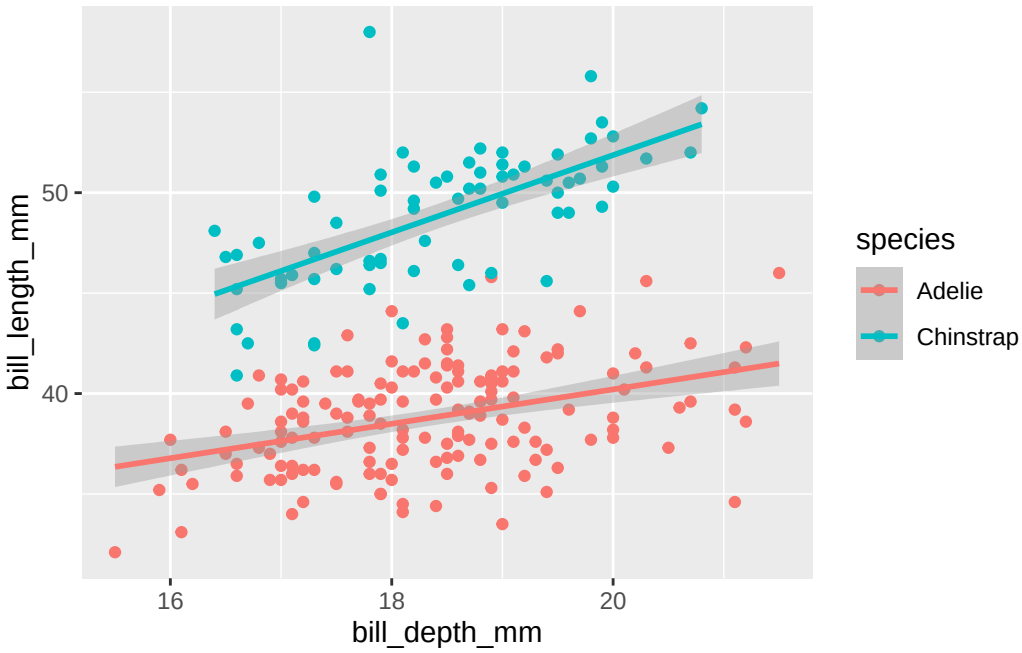
Q2.1 Make a graph

Graphing bill length on the y and bill depth on the x with point colored by species

```r
ggplot(penguins.AC, aes(x= bill_depth_mm, y = bill_length_mm, color = species)) +
  geom_point() +
  geom_smooth(method = "lm")
```

```
`geom_smooth()` using formula = 'y ~ x'

Warning: Removed 1 row containing non-finite outside the scale range
(`stat_smooth()`).
```

```
Warning: Removed 1 row containing missing values or values outside the scale range
(`geom_point()`).
```



Q2.2 Set up and run an *interactive* model

```
# bill length by bill depth and species - INTERACTIVE model
m.bill.spp.interactive <-
  brm(data = penguins.AC, # Give the model the penguins data
      # Choose a gaussian (normal) distribution
      family = gaussian,
      # Specify the model here.
      # First, we write the equation with a as an intercept and b as a slope next to bill dep
      bf(bill_length_mm ~ 0 + a + b*bill_depth_mm,
         # Then, we specify that we want our intercept, a, to vary with species
         a ~ 0 + species,
         # Next, we specify that we want our slope, b, to ALSO vary with species
         # (this is the interaction part!!)
         b ~ 0 + species,
         # Lastly, we tell it that we are writing in a particular notation
         nl = TRUE),
      # Here's where you specify parameters for executing the Markov chains
```

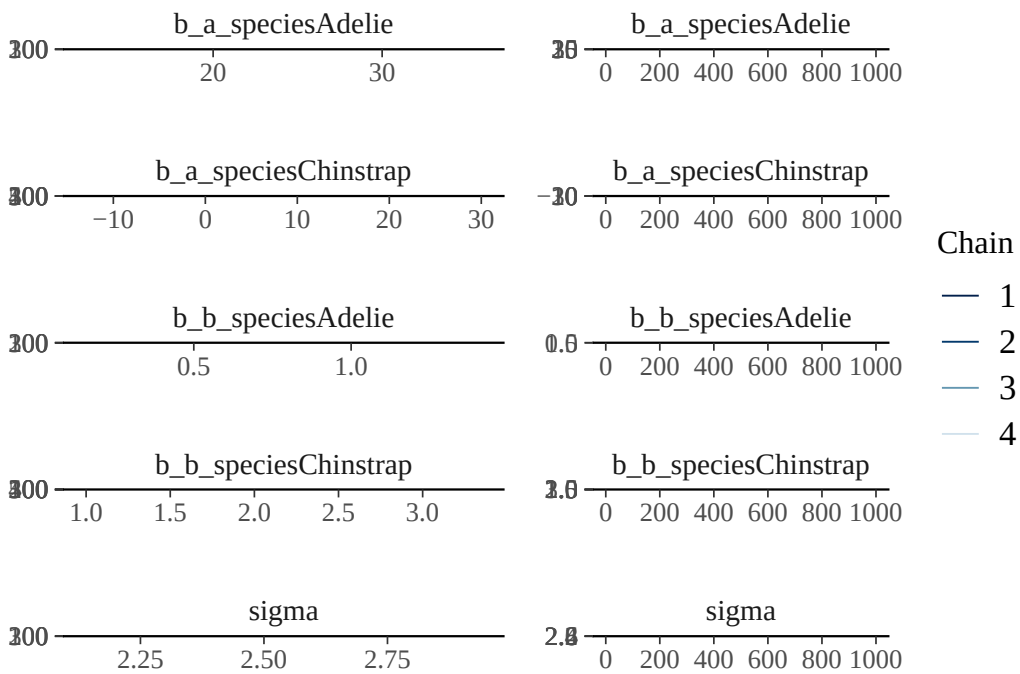
```r
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Setting the "seed" determines which random numbers will get sampled.
      # In this case, it makes the randomness of the Markov chain runs reproducible
      # (so that both of us get the exact same results when running the model)
      seed = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.bill.spp.interactive")
```

Q2.3 Assess the model

```r
summary(m.bill.spp.interactive)
```

```
 Family: gaussian
  Links: mu = identity
Formula: bill_length_mm ~ 0 + a + b * bill_depth_mm
         a ~ 0 + species
         b ~ 0 + species
   Data: penguins.AC (Number of observations: 219)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
                   Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
a_speciesAdelie       23.01      3.12    16.83    28.92 1.00     1670     1811
a_speciesChinstrap    13.47      4.90     4.09    22.92 1.00     1785     1446
b_speciesAdelie        0.86      0.17     0.54     1.20 1.00     1675     1692
b_speciesChinstrap     1.92      0.27     1.41     2.43 1.00     1790     1465

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sigma     2.50      0.12     2.28     2.74 1.00     2318     2241

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m.bill.spp.interactive)
```



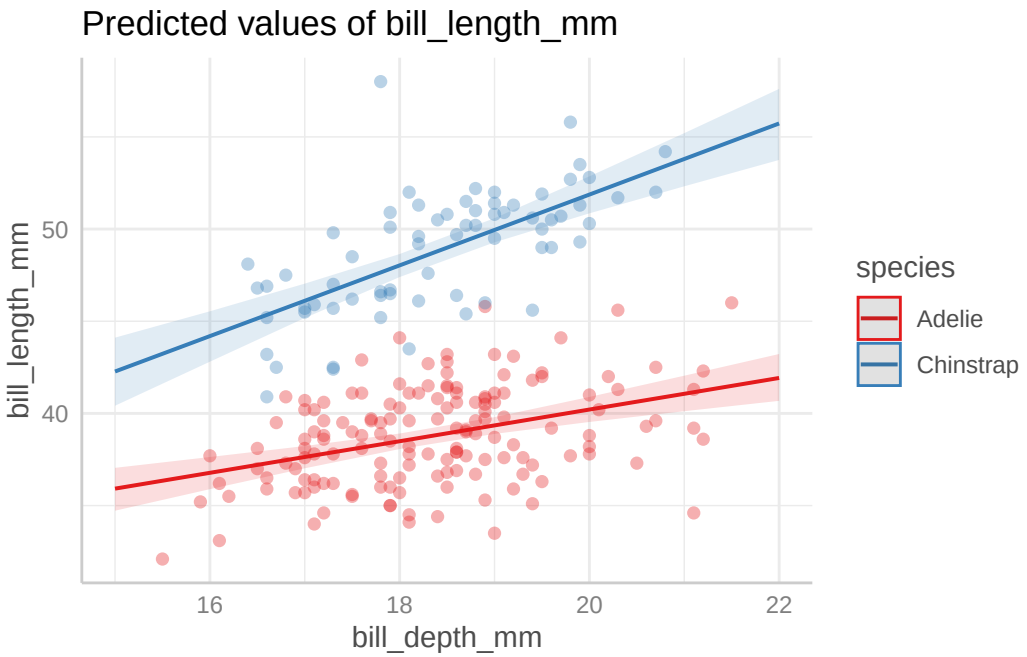
Looks like the model ran correctly. Rhat is 1, the posterior distributions are normal, and the chains are overlapping.

Q2.4 Plot model predictions

Plot the model predictions using `predict_response()`

```
preds.add <- predict_response(m.bill.spp.interactive,
                              terms = c("bill_depth_mm", "species"))

plot(preds.add, show_data = TRUE)
```

```
Data points may overlap. Use the `jitter` argument to add some amount of
  random variation to the location of data points and avoid overplotting.
```



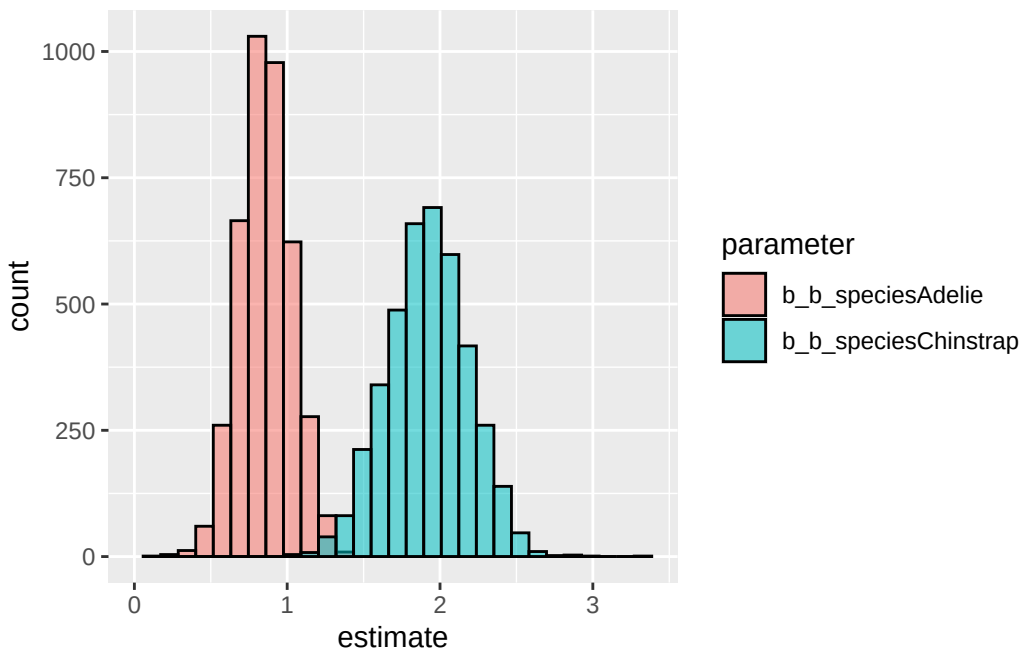
Q2.5 Plot posteriors

```
# Store posterior parameter estimates
model_posterior_samples <- as_draws_df(m.bill.spp.interactive)

# Wrangle and then plot samples for the slopes (b) specifically
model_posterior_samples %>%
  # Select the columns we care about (exclude the posterior estimates for the intercepts and
  dplyr::select(b_b_speciesAdelie, b_b_speciesChinstrap) %>%
  # Pivot to long format for easy ggplotting
  pivot_longer(cols = everything(),
               names_to = "parameter",
               values_to = "estimate") %>%
  # Plot as a two overlapping histograms, with the fill colored by parameter
  ggplot(aes(x = estimate, fill = parameter)) +
  geom_histogram(alpha = 0.55, color = "black",
                 # position = "identity" makes the histograms overlay each other instead of s
                 position = "identity") +
  xlim(0, NA)
```

```
Warning: Dropping 'draws_df' class as required metadata was removed.
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_bar()`).
```



Q2.6 Interpret the model

Interpret your model by answering:

1. How does bill depth influence bill length, and how does it differ by species? Remember to describe the effect using the units to make it biologically meaningful.

   Bill depth has a positive effect on bill length for both species. For Adelie, for every millimeter increase in bill depth there is a 0.86 increase in bill length. For Chinstrap, there is a 1.92mm increase in bill length for every millimeter of bill depth.

2. Does it seem like the slope estimates are different from zero? Why?

   Yes, we can conclude that all slope estimates are different from zero since the 95% credibility intervals don't include zero.

3. Does it seem like the slope estimates are different from one another (e.g. do Adelie and Chinstrap penguins have different slopes)? Why?

   Yes, the slopes estimates appear different from each other since the plot of the posteriors only overlaps slightly.

Q2.7 Which model has better predictive power?

```
#Running an additive model to compare
# flipper and body mass model
m.bill.spp.additive <-
  brm(data = penguins.AC, # Give the model the penguin data
      # Choose a gaussian (normal) distribution
      family = gaussian,
      # Specify the model here.
      bill_length_mm ~ 0 + species + bill_depth_mm,
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Setting the "seed" determines which random numbers will get sampled.
      # In this case, it makes the randomness of the Markov chain runs reproducible
      # (so that both of us get the exact same results when running the model)
      seed = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.bill.spp.additive")
```

```
#PSIS additive model
loo(m.bill.spp.additive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

         Estimate   SE
elpd_loo   -518.2 11.9
p_loo         4.4  0.9
looic      1036.3 23.7
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.2, 1.1]).

All Pareto k estimates are good (k < 0.7).
```

```
See help('pareto-k-diagnostic') for details.
```

```r
#PSIS interactive model
loo(m.bill.spp.interactive)
```

```
Computed from 4000 by 219 log-likelihood matrix.

         Estimate   SE
elpd_loo   -513.4 12.5
p_loo         5.2  1.0
looic      1026.7 24.9
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.1]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
#WAIC additive model
waic(m.bill.spp.additive)
```

```
Warning:
1 (0.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.


Computed from 4000 by 219 log-likelihood matrix.

          Estimate   SE
elpd_waic   -518.2 11.9
p_waic         4.4  0.9
waic        1036.3 23.7

1 (0.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```r
#WAIC interactive model
waic(m.bill.spp.interactive)
```

```
Warning:
1 (0.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```
Computed from 4000 by 219 log-likelihood matrix.

          Estimate   SE
elpd_waic   -513.4 12.4
p_waic         5.2  1.0
waic        1026.7 24.9

1 (0.5%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

The interactive model has lower WAIC and PSIS so therefore has higher predictive power.